

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions	
CLO1	Familiarisation with the database system, RDBMS, SQL, transaction processing and concurrency control.
CLO2	Design and development of a database application system.

Letter Symbols Meaning	Bloom's Levels of the Questions					
	R Remember	U Understand	Ap Apply	An Analyze	E Evaluate	C Create

- | | | | | | | | | | | | | | | | | | | | |
|--|--|------|---|---|------|---|---|------|---|---|------|----|---|------|---|---|------|----|--|
| <p>Q1. a) Explain any four limitations of file-based systems. Illustrate your answer with brief examples.</p> <p>b) Explain two-tier and three-tier database architectures with suitable diagrams</p> <p>c) Consider the relation:
Employee (EmpID, Name, DeptName, Salary)
Distinguish between physical schema, logical schema, and instance with respect to the above relation.</p> <p>Q2. a) Consider the following relations for University management System Database:
Instructor (ID, Name, DeptName, Salary)
Teaches (ID, CourseID, Semester, Year)
Student (ID, Name, DeptName, Email)
Construct Schema diagram considering appropriate primary key and foreign key.</p> <p>b) Consider the relation:
Student (ID, Name, DeptName, Email)
i. Identify a candidate key for the relation. Justify your answer.
ii. Distinguish between primary key and foreign key with suitable examples.</p> | <table border="0"> <tr> <td>CLO1</td> <td>U</td> <td>4</td> </tr> <tr> <td>CLO1</td> <td>U</td> <td>3</td> </tr> <tr> <td>CLO1</td> <td>U</td> <td>3</td> </tr> <tr> <td>CLO1</td> <td>Ap</td> <td>5</td> </tr> <tr> <td>CLO1</td> <td>U</td> <td>5</td> </tr> <tr> <td>CLO1</td> <td>Ap</td> <td></td> </tr> </table> | CLO1 | U | 4 | CLO1 | U | 3 | CLO1 | U | 3 | CLO1 | Ap | 5 | CLO1 | U | 5 | CLO1 | Ap | |
| CLO1 | U | 4 | | | | | | | | | | | | | | | | | |
| CLO1 | U | 3 | | | | | | | | | | | | | | | | | |
| CLO1 | U | 3 | | | | | | | | | | | | | | | | | |
| CLO1 | Ap | 5 | | | | | | | | | | | | | | | | | |
| CLO1 | U | 5 | | | | | | | | | | | | | | | | | |
| CLO1 | Ap | | | | | | | | | | | | | | | | | | |

Q3. Consider the following table "Payment". CLO2 Ap 10

Payment_Id	Student_id	Amount	Status	Payment_date
1	1001	40000	Paid	8.02.26
2	1003	45000	Paid	8.02.26
3	1027	40000	Due	-

- Answer**
- the following question
- i. Write an SQL query to create this table assuming Payment_Id as primary key.
 - ii. Write an SQL query to insert the given data into the table.
 - iii. Write an SQL query to show the students whose payment is not due.

- iv. Find total paid amount from this table.
- v. Change the status of student having Id 1027 to "Paid" and add date '11.3.26' in this table.

OR

- | | | | | | | | |
|---|---|------|----|---|------|----|---|
| <p>Q3. a) Consider the following 'Hostel Management System' database:
Students (Student_id, Student_name, Department, Room_no, Admission_date, Fee_status)
Rooms (Room_no, Capacity, Room_type, Monthly_rent)
Payments (Payment_id, Student_id, Amount, Payment_date, Status)
i. Find the rooms with capacity greater than 2.
ii. Find the name and department of students whose payment status is 'Due'.
iii. Find total number of students in each department.</p> <p>b) Explain basic SQL structure. Write difference between Drop and delete command in SQL..</p> | <table border="0"> <tr> <td>CLO2</td> <td>Ap</td> <td>5</td> </tr> <tr> <td>CLO2</td> <td>An</td> <td>5</td> </tr> </table> | CLO2 | Ap | 5 | CLO2 | An | 5 |
| CLO2 | Ap | 5 | | | | | |
| CLO2 | An | 5 | | | | | |

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.sc (Engg.)
Course Code: CCE-3505
Total Marks: 30

Semester: Spring-2025
Course Title: Database Management Systems
Time: 1 Hours 30 Minutes

(I) Answer all the questions. The figures in the right-hand margin indicate full marks.

(II) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1 Familiarisation with the database system, RDBMS, SQL, transaction processing and concurrency control.

CLO2 Design and development of a database application system.

Bloom's Levels of the Questions

Letter Symbols
Meaning

R
Remember

U
Understand

Ap
Apply

An
Analyze

E
Evaluate

C
Create

- Q1.**
- a) Define DBMS. What are the key applications of database systems in real-world scenarios? Explain with proper examples. **CLO1 R 4**
 - b) What is RDBMS? How does a database system improve efficiency compared to a file-based system? **CLO1 U 4**
 - c) Explain Query Processor Components of DBMS. **CLO1 R 2**

Q2. Student (Student_Id, S_Name, CGPA)
Teacher (Teacher_Id, T_Name, DOB, Department)
Course (C_Id, C_Title, Credits, Teacher_Id)
Enrollment (E_Id, Student_Id, C_Id, Grade)
Consider the above Student Course Management database

- a) Define superkey, primary key and foreign key. **CLO1 R 3**
- b) What are the appropriate primary keys and foreign keys for each table from the given Student Course Management database? **CLO2 An 2**
- c) What is database schema and instance? Construct a schema diagram for the above Student Course Management database. **CLO2 Ap 5**

Q3. Consider the following table "Registration".

CLO2 Ap 10

Id	Name	Course	Semester	Status	Deadline
1001	Karim	DBMS	Spring 25	Active	8.02.25
1002	Rahim	DS	Spring 25	Active	8.02.25
1003	Rahman	EEE	Spring 25	Pending	10.03.25

Answer the following question

- i. Write an SQL query to create this table.
- ii. Write an SQL query to insert the given data into the table.
- iii. Write an SQL query to Show the students whose registration status is "Active".
- iv. Change the status of Rahman to "Active" in this table.
- v. Find out how many records are present in this table.

OR

Q3. Consider the following table "Employee".

CLO2 Ap 10

Emp Id	Emp Name	Designation	Salary
1	Salam	Sr. Developer	50000
2	Razzak	Developer	35000
3	Azim	Trainee	20000
4	Ahad	Trainee	22000

- i. Write an SQL query to create this table.
- ii. Write an SQL query to insert the given data into the table.
- iii. Write an SQL query to show the employees whose salary is more than 20000 and whose designation is "Trainee".
- iv. What is an aggregate function in SQL? Give some examples of aggregate functions.
- v. Explain different data types in SQL.

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.sc (Engg.)
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Semester: Autumn- 2025
 Course Title: Database Management Systems
 Time: 1 Hours 30 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1** Familiarisation with the database system, RDBMS, SQL, transaction processing and concurrency control.
CLO2 Design and development of a database application system.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- Q1. a)** a) An organization currently maintains employee records using traditional file storage. Explain how adopting a database would improve data management, highlighting the key elements of a database. Illustrate, with examples, how a DBMS helps reduce data inconsistency compared to simple file systems. CLO1 R 5
- b)** b) Explain the difference between a DBMS and an RDBMS using a simple example, such as storing student information. In your explanation, describe how data is organized and stored in a relational database. Describe the query processor architecture in a relational database. CLO1 U 5

- Q2. a)** What is schema and instance? Define primary key, foreign key and composite key. CLO2 R 5
- b)** Consider the following 'Transport Management System' database- CLO2 An 5
Vehicles(Vehicle_id, Type, Driver_id, Purchasing_date, Price, Capacity, Fitness_certificate_no)
Vehicle_Fitness(Fitness_certificate_no, Expiry_date, Inspection_date)
Drivers(Driver_id, Driver_name, Licence_no, Salary, Joining_date)

Construct a schema diagram for the above 'Transport Management System' considering appropriate primary keys and foreign keys.

SUM, MAX, MIN, COUNT()

- Q3. a)** What are aggregate functions? Name at least two and explain their use. Give a scenario for when you would use DELETE and when you would use DROP in SQL. CLO2 AP 5
- b)** Consider the following table "Students". CLO2 Ap 5

Id	Name	Department	Semester	Status	GPA
1	Tanvir	CSE	Spring 25	Active	3.60
2	Laila	CCE	Fall 24	Active	3.40
3	Farhan	BBA	Spring 25	Pending	3.10
4	Nabila	CSE	Fall 24	Active	3.85
5	Arif	CCE	Spring 25	Active	3.75

Answer the following questions

- i. Write an SQL query to create this table.
- ii. Write an SQL query to calculate the average GPA of all students. Display the result.
- iii. Write an SQL query to display the number of students in each department. Show the output in a tabular form.

OR

- Q3. a)** **Vehicles**(Vehicle_id, Type, Driver_id, Purchasing_date, Price, Capacity, Fitness_certificate_no) **CLO2 Ap 5**
Vehicle_Fitness(Fitness_certificate_no, Expiry_date, Inspection_date)
Drivers(Driver_id, Driver_name, Licence_no, Salary, Joining_date)

Consider the above 'Transport Management System' Schema and write SQL for the following-

1. Find the number of cars.
 2. Find the vehicles whose fitness certification will expire before '2024-11-2'.
 3. Find the drivers who joined after 2018 and Salary is more than 20000 .
- b)** What is DML? Write basic query structure and explain each clause. **CLO2 Ap 5**

100

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Class Test

Program: B.sc (Engg.)

Course Code: CCE-3505

Total Marks: 15

Semester: Autumn 2024

Course Title: Database Management Systems (DBMS)

Time: 45 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

- CLO1 Introduction to database and its languages
CLO2 Understanding ERD and mapping to database
CLO3 Understand and implement SQL expression / query

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Answer any <u>Three</u> of the following Questions.						
Q1.	a)	Define any 2 - i) Weak entity ii) Composite attribute iii) Multivalued attribute iv) Derived attribute v) Many-to-many relationship	CLO2	R	2	
	b)	What is Generalization in database design, and under what circumstances should you use Generalization instead of Specialization? Provide an example to illustrate your explanation.	CLO2	U	3	
OR (b/c)						
	c)	What is Specialization in database design, and under what circumstances should you use Specialization instead of Generalization? Provide an example to illustrate your explanation.	CLO2	U	3	
Q2.	d)	Suppose you have graduated from International Islamic University Chittagong and recently joined a reputable software firm as a Junior Software Engineer. For your first assignment, you are tasked with designing and developing a database for a management system. Now complete the following task Task 1 or Task 2 Task 1: Design an ER Diagram i) Identify and list the main entities required for the management system (any). ii) Define the attributes for each entity. iii) Establish the relationships between the entities. iv) Draw the ER diagram showing the entities, attributes, and relationships, ensuring cardinality is represented Task 2: Map the ER Diagram to a Relational Database Model/ Database i) Convert the entities into database tables. ii) Specify the primary keys for each table. iii) Define the foreign keys to maintain relationships between the tables. iv) Ensure all attributes are mapped as columns with appropriate data types	CLO2	Ap	5	

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: B.sc (Engg.)
Course Code: CCE-3505
Total Marks: 30

Semester: Spring-2024 / Autumn- 2023
Course Title: Database Management Systems
Time: 1 Hours 30 Minutes

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions								
CLO1	Familiarisation with the database system, RDBMS, SQL, transaction processing and concurrency control.							
CLO2	Design and development of a database application system.							
Bloom's Levels of the Questions								
Letter Symbols		R	U	Ap	An	E	C	
Meaning		Remember	Understand	Apply	Analyze	Evaluate	Create	
Q1.	a)	What is Data Model? Describe the main categories of the data models?				CLO1	R	3
	b)	What are four significant differences between a file-processing system and a database system.				CLO1	U	3
	c)	What are DDL and DML? Give examples for both.				CLO1	U	4
Q2.	Customer (CustomerId, Name, Email, PhoneNumber, Address) Room (RoomId, RoomNumber, RoomType, Price, Capacity, Description) Booking (BookingId, CustomerId, RoomId, CheckInDate, CheckOutDate, TotalPrice, PaymentStatus)							
	a)	Define superkey, primary key and foreign key.				CLO1	R	3
	b)	Consider the above room booking database. What are the appropriate primary keys and foreign keys?				CLO2	An	2
	c)	Construct a schema diagram for the above room booking database.				CLO2	Ap	5
Q3.	Consider the following Library database- Book (BookId, Title, Author, Genre, PublicationDate, Publisher, ISBN) Member (MemberId, Name, Address, Email, PhoneNumber) Borrowing (BorrowingId, MemberId, BookId, BorrowDate, ReturnDate) Note: Borrowing schema contains two references from Book and Member relations.							
	a)	Write SQL statements to create Book, Member and Borrowing tables.				CLO2	Ap	5
	b)	Write an SQL query to insert a new record into the "Book" table for a book titled "The Great Gatsby" by F. Scott Fitzgerald, published on January 1, 1925, under the genre "Classic", published by "Scribner" and ISBN is 9780134763415.				CLO2	Ap	2.5
	c)	Write an SQL query to update the "PhoneNumber" column from "01678344556" to "01726877878" in the "Member" table for a specific member whose MemberId is "234".				CLO2	Ap	2.5
OR								
Q3.	Consider the following Hospital Database- Patient (PatientId, PatientName, Address, PhoneNumber, Email) Doctor (DoctorId, DoctorName, Specialization, Experience) Appointment (AppointmentId, DoctorId, PatientId, AppointmentDate)							
	a)	Write an SQL query to join the Appointment table with the Patient and Doctor tables to retrieve the names of patients and doctors along with their appointment dates . Note that you need to combine all three tables.				CLO2	Ap	3
	b)	What is transaction in DBMS? What do you know about ACID property? What does A mean in ACID- explain.				CLO2	An	4
	c)	Explain referential integrity with example.				CLO2	An	3

International Islamic University Chittagong (IIUC)

Department of Computer and Communication Engineering

Midterm Examination

Program: **B.Sc. (Engg.)**
 Course Code: **CCE-3505**
 Total Marks: **30**

Semester: **Autumn 2020**
 Course Title: **Database Management Systems**
 Time: **1 Hours 30 Minutes**

- (I) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (II) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions								
CLO1	Familiarization with the database system, RDBMS, SQL, transaction processing and concurrency control.							
CLO2	Design and development of a database application system							
Bloom's Levels of the Questions								
Letter Symbols	Meaning	R	U	Ap	An	E	C	
		Remember	Understand	Apply	Analyze	Evaluate	Create	
Q1.	a)	Define database and RDBMS. Write the advantages of DBMS. Differentiate between File System and DBMS				CLO1	R	5
	b)	Define entity and attribute. Explain the types of DBMS architecture.				CLO1	R	5
Q2.	a)	What is primary key and foreign key? Explain ACID properties in DBMS with examples				CLO1	Un	5
	b)	Define data model schema and relationship. Describe the types relationship with examples.				CLO1	Un	5
Q3.	a)	Suppose you are given the following requirements for a simple database for the IIUC Football League (IFL): the IFL has many teams, - Each team has a name, a city, a coach, a captain, and a set of players, - Each player belongs to only one team, - Each player has a name, age, a position, a skill level, and a set of injury records, - A team captain is also a player, - A game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 1999) and a score (such as 4 to 2). Construct a clean and concise ER diagram for the IFL database				CLO1	Un/Ap	10
	b)	Write an expression in SQL for the following queries: i) Create a table called "Persons" that contains five columns: PersonID, LastName, FirstName, Address, and City ii) Insert 3 new records in the "Persons" table. iii) Delete the "city" column from the "Persons" table. iv) Delete a record from the "Persons" table.				CLO1	Un/I	10

OR

Q3. a) Create an ER diagram for each of the following descriptions:

CLO1

- I. Each company operates four departments, and each department belongs to one company.
- II. Each department in part (I) employs one or more employees, and each employee works for one department.
- III. Each of the employees in part (II) may or may not have one or more dependants, and each dependant belongs to one employee.
- IV. Each employee in part (III) may or may not have an employment history.
- V. Represent all the ER diagrams described in (I), (II), (III), and (IV) as a single ER diagram.

b) Consider the following 'Employee' table:

CLO1

Emp_id	Name	Address	Salary
1	Aziz	Chattogram	28000
2	Abrar	Dhaka	25000
3	Azad	Sylhet	20000

Write a SQL query to rename the table.

Write a SQL query to display employee name, whose salary > 21000.

Differentiate between DELETE and TRUNCATE statements.

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG (IIUC)
DEPARTMENT OF COMPUTER & COMMUNICATION ENGINEERING
MIDTERM EXAMINATION

Program: BSc (Engg.)
Course Code: CCE-3505
Total Marks: 30

Semester: Autumn 2022
Course Title: Database Management Systems
Time: 1 Hours 30 Minutes

Answer all questions. The figures in the right-hand margin indicate full marks. Course Outcomes (COs) and Bloom's Levels (BLs) are mentioned in each question.

Course Outcomes (COs) of the Questions						
CO1	Understand the necessity and efficient strategies for managing large datasets.					
CO2	Understand the mathematical foundations and apply programmatic techniques to manage databases.					
Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Question 1: (5+3+2) marks – CO: [CO1, CO2] – BL: [R, U, An]

- Discuss the main characteristics of the database approach and how it differs from traditional file systems.
- What is the difference between a key and a superkey? Why do we designate one of the candidate keys of a relation to be the primary key? Explain with a suitable example.
- Give examples of systems in which it may make sense to use traditional file processing instead of a database approach.

Question 2: (2.5+2.5+2.5+2.5) marks – CO: [CO1, CO2] – BL: [U, Ap, C]

employee (person_name, street, city)
works (person_name, company_name, salary)
company (company_name, city)

Consider the given Employee database. Give an expression in the relational algebra and a SQL statement to express each of the following queries:

- Find the name of each employee who lives in city "Chittagong".
- Find the ID and name of each employee who does not work for "BigBank".
- Find the ID, name, street address, and city of residence of each employee who works for "Big-Bank" and earns more than \$10000.
- Find the ID and name of each employee in this database who lives in the same city as the company for which she or he works.

Please turn over ...

Question 3: (2+4+4) marks – CO: [CO1, CO2] – BL: [R, U, Ap, C]

Consider the following schema:

Department(id, name, address)

Faculty(id, dept_id, first_name, last_name, join_year)

Course(id, dept_id, title, credits)

Student(id, dept_id, first_name, last_name)

Teaching(faculty_id, course_id, sem)

Registration(student_id, course_id, sem)

- (a) What is the relational algebraic operation do you use to combine more than one relations.
- (b) Write a JOIN query in SQL to select all the students registered for the fifth semester. Note that you need to combine Registration and Student tables.
- (c) Write a JOIN query in SQL to select all the faculties who works for "CCE" department. You also need to include all faculties who are not working for any department. Note that you need to combine Faculty and Department tables and select appropriate join form left, right, or full join.

OR

Question 3: (5+2.5+2.5) marks – CO: [CO1, CO2] – BL: [R, U, Ap, C]

Three relational schema are given below.

Patient(ssn:string, name:string, address:string, age:integer)

Doctor(ssn:string, name:string, specialty:string, experience:integer)

PrimaryPhysician(patientSSN:string, doctorSSN:string, firstAppointment:date)

Note that the primary physician schema contains two references from Patient and Doctor relations and SSN should be unique in each relation.

- (a) Write SQL statements to create Patient, Doctor, and Primary Physician tables.
- (b) Set the specialty of all doctors to 'Cardiologist' whose name starts with 'Adam'
- (c) Delete all the patient from the 'Patient' table whose age is less than one year.